Yang Wu

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Research Interests

Imagination and Mental Simulation, with a focus on the role of memory recombination and memory retrieval.

Education

Soochow University 2023 – 2027
B.S. in Mathematics & B.A. in Music Composition *Taipei City, Taiwan*

Neuropsychiatric Disorder Summer Associate School (IBRO) 2024
National Taiwan University *Taipei City, Taiwan*

Cognitive Neuroscience Summer School 2024
National Central University *Taoyuan City, Taiwan*

Florida State University 2022 – 2023
B.A. in Music (Piano Performance) *Tallahassee, FL, USA*

Research Experience

Kanaka Rajan lab, Harvard University 2026/2 – Present
Research Assistant, Harvard Medical School / Kempner Institute *Boston, MA*
Advisor: John Vastola

- **Characteristics of Multi-Timescale Efficient-Coding Neural Network under Energy Efficiency Constraint:** Designed and ran Contrastive Predictive Coding experiments with InfoNCE as the Mutual Information in the Efficient-Coding hypothesis, confirming three theoretical predictions about how representation timescales emerge in Efficient-Coding networks with temporal smoothness energy costs.
- **Computational Model of Molecular Memory:** Developing computational model of memory storage at the molecular level, inspired by "The molecular memory code and synaptic plasticity" (Gershman et al., 2023); building neural-symbolic architecture using Combinatory Logic primitives within neurons, inspired by "An RNA-Based Theory of Natural Universal Computation" (Akhlaghpour et al., 2022).

James Whittington lab, University of Oxford 2025/7 – Present
Research Assistant, Department of Experimental Psychology *Oxford, UK*
Advisors: James Whittington & Joseph Warren

- **Cognitive Map Reorganization in Sequential Memory Task:** Identified specific algorithms the RNN uses in sequential memory task, extending previous work "A tale of two algorithms" (Whittington et al., 2024) to model how the hippocampus reorganizes its cognitive map during sequence learning.
- **Neural Subspace Stretching as a Representational Signature of Sequence Learning in RNN:** Used linear decoders to test competing hypotheses about observation encoding in RNN. Identified the "stretching" of neural subspaces during sequence learning of RNN that generalized across multiple loop lengths, and quantified different learning phases' decoder alignment via cosine similarity.

Marcelo Mattar lab, New York University 2025/7 – Present
Research Assistant, Department of Psychology *New York, NY*
Advisor: Marcelo Mattar

- **Modeling Episodic Simulation through Memory Recombination:** Built key-value episodic memory module integrated with GRU controller to model human episodic simulation, i.e. how humans recombine episodic memories from the past to implement future planning or imagination (Schacter et al., 2007).
- Designed three-phase reinforcement learning task (exploration → mental simulation → execution) to study how episodic memories from past experiences support future planning and imagination.
- Established that Phase 2 mental simulation improves Phase 3 execution; investigating whether this benefit reflects explicit sequential plan formation during Phase 2 mental simulation or mere enrichment of key-value memory representations.

Kanaka Rajan lab, Harvard University 2025/7 – 2025/12
Research Assistant, Harvard Medical School / Kempner Institute *Boston, MA*
Advisors: Ryan Badman & John Vastola

- **Interpreting Planning-Like Behavior in Naturalistic Foraging RL Agents:** Analyzed emergent planning-like behaviors ethologically in model-free reinforcement learning agents within naturalistic environments.
- Implemented Sparse Autoencoder, Generalized Linear Model, Decoder and Auto-Correlation to probe agents' internal representations and external behavior.
- Contributed code and analysis to *Deep RL Needs Deep Behavior Analysis: Exploring Implicit Planning by Model-Free Agents in Open-Ended Environments*, *NeurIPS 2025*; formally recognized in acknowledgements.

Yu-Wei Wu lab, Academia Sinica 2024/7 – 2025/7
Research Assistant, Institute of Molecular Biology *Taipei, Taiwan*
Advisors: Yu-Wei Wu & Yiko Chen

- **Decomposing Inter-Regional Neural Dynamics during Mouse Motor Control:** Used Current-Based Decomposition Method (CURBD) and Dissociative Prioritized Analysis of Dynamics (DPAD) to model and describe inter-regional communication between M1, M2 and S1 during mouse on-cue reaching task.

Yu-Te Wang lab, Academia Sinica 2024/2 – 2025/3
Research Assistant, Research Center for Information Technology Innovation *Taipei, Taiwan*
Advisor: Yu-Te Wang

- **Development of 3D Printed Semi-Dry EEG Electrodes:** Built EEG electrodes with 3D printing; collected human EMG/EEG data; applied CompactCNN for EEG analysis. Led to multiple conference and IEEE journal publications.
- VR/AR-Based HCI Telerehabilitation Systems for EMG Motor Recovery using Unity.

Publications

3D printed watermill-like semi-dry electrodes for BCI applications. IEEE Transactions on Neural Systems and Rehabilitation Engineering, 2025.
C.-M. Chung, Y.-T. Wang*, J.-B. Lu, Y.-C. Hsu, Y.-J. Su, **Y. Wu**, C.-F. Hung

Posters

HCI Technologies in rehabilitation: enhancing physical therapy with sEMG, VR and LLM-Generated music. Society for Neuroscience Annual Meeting, 2024.
Y. Wu*, Y.-J. Su, J.-B. Lu, Y.-T. Wang*, T.-H. Cheong

RehabVerse: Mixed Reality Solution for Telerehabilitation. National Biotechnology Research Park Pitch Day, 2024.
Y. Wu*, Y.-J. Su, J.-B. Lu, Y.-H. Chen, Y.-T. Wang*

BrainPrint: Innovative Head-Mounted EEG Technology for Secure Personal Identification. IEEE Brain Discovery & Neurotechnology Workshop, 2024.
Y.-L. Chu, C.-C. Tsao, C.-M. Chung, C.-F. Hung, Y.-Y. Lee, J.-B. Lu, **Y. Wu**, Y.-H. Chen, Y.-T. Wang*

Awards, Grants & Extracurricular

Ministry of Education Study Abroad Programme for Future Scholars in Humanities and Social Sciences (US$50,000)	2026
Computational and Systems Neuroscience Conference (COSYNE) Undergraduate Travel Grant (US$1,500)	2025
Chu-Chin Hsu Scholar	2025
Dean's List (Academic Excellence Award) × 2	2025
Distinction Award, International ICT Innovation Services Award (BrainPrint)	2024
National Science and Technology Council Neuroscience Camp	2024
Dean's List (Academic Excellence Award)	2024
Academia Sinica Statistical Science Camp	2024
Finalist, National Biomedical Engineering Award (BrainPrint)	2024

Courses & Skills

Selected Courses (grades on 100-point scale): **Computational Neuroscience (Grad Level) 95**, **Statistical Mechanics (Grad Level) 87**, Advanced Calculus (Real Analysis) 99, Linear Algebra 96, Programming in Life Science 97, Life Science (Physiology) 95.

Skills: Python; Bayesian inference; Deep learning (PyTorch, RNNs, Transformers); Reinforcement learning (model-free and model-based, PPO); Statistical modeling (GLMs, sparse autoencoders, linear decoders); Data collection (EEG, EMG); Tools (Git, Blender, MATLAB, 3D printing).

Music

Milonga – Latin-American piece for Cello and Piano. vimeo.com/1036267350

Zhang Beihai said good-bye to his father – Solo Piano. vimeo.com/1036267291

Weathering With You Movie Trailer (Re-score). vimeo.com/1036267308